NLP

Language

- Language of mathematics and/or logic
- Programming language
- Natural language

What will happen if we communicate in a formal language?

Language: Formal and Natural

a language is a set of strings defined over some alphabet

- alphabet
- word
- vocabulary and dictionary
- sentence
- segment (paragraph)
- document

Can you think of data structure(s) for a sentence/segment/document?

Inverse problem

- Find a model that describes how a given set of strings were generated
  - Date expressions
  - Email-ids

Regular Language

- a language recognized by a finite automaton/regular expression

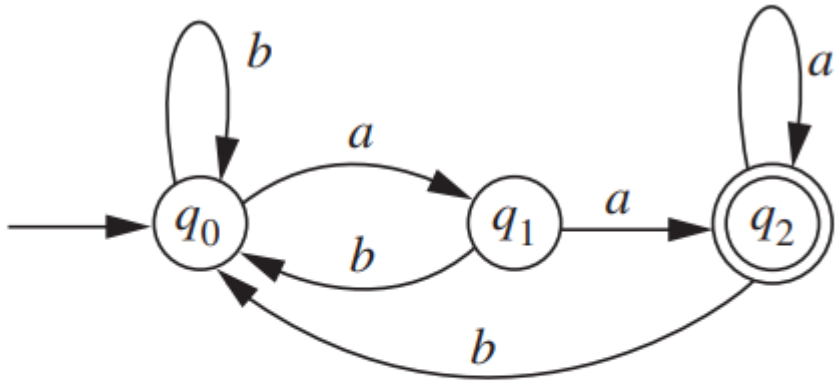

Figure 2.2 |
An FA accepting the strings ending with aa.

Regular expression: $(a+b)^*aa$

Grammar:

$$S \rightarrow Paa$$
$$P \rightarrow aP \mid bP \mid \wedge$$

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Finite Automaton

Definition 2.11 A Finite Automaton

A *finite automaton* (FA) is a 5-tuple $(Q, \Sigma, q_0, A, \delta)$, where

Q is a finite set of *states*;
Σ is a finite *input alphabet*;
$q_0 \in Q$ is the *initial* state;
$A \subseteq Q$ is the set of *accepting* states;
$\delta : Q \times \Sigma \to Q$ is the *transition* function.

For any element q of Q and any symbol $\sigma \in \Sigma$, we interpret $\delta(q, \sigma)$ as the state to which the FA moves, if it is in state q and receives the input σ.

- Finite automaton and regular expression for strings of **even** length

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Regular Expression (RegEX)

Character	Description	Example
[]	A set of characters	"[a-m]"
\	Signals a special sequence (can also be used to escape special characters)	"\d"
.	Any character (except newline character)	"he..o"
^	Starts with	"^hello"
$	Ends with	"planet$"
*	Zero or more occurrences	"he.*o"
+	One or more occurrences	"he.+o"
?	Zero or one occurrences	"he.?o"
{}	Exactly the specified number of occurrences	"he.{2}o"
|	Either or	"falls|stays"
()	Capture and group	

Special Sequences

- \b: **\b**york**\b**
- \d: +91**\d{5}**-**\d{5}**
- \s: IIT**\s**Palakkad
- \w: [a-z0-9_]
- \B, \D, \S, \W
- a|e|i|o|u: [aeiou]~[^aeiou]

Find roll number from a student's email address:

^(\d+)@

https://www.w3schools.com/python/python_regex.asp

RegEx for email address

[a-zA-Z0-9][a-zA-Z\._0-9]+@([a-zA-Z0-9]+\.){1,3}[a-z]+

Reading Assignment

- Evaluating Cognitive Capacities in AI Systems by Melanie Mitchell
- LLMs are Cultural and Social Technologies by Alison Gopnik
  - https://www.youtube.com/watch?v=yKbeCvOKvMc

RegEx for Date

- (?<Day>\d{1,2})[/-](?<Month>\d{1,2})[/-](?<Year>(?:\d{4}|\d{2}))

- (?<Day>\d{1,2})[/-](?<Month>**(0?[1-9])|(1[0-2])**)[/-](?<Year>(?:\d{4}|\d{2}))

- 15-August-1947
- 15-Aug-1947
- 15/08/1947
- 08/15/1947
- 15th of August, 1947
- fifteen august nineteen forty-seven

- Palindromes

Context Free Languages

- Palindromes

$$S \rightarrow \Lambda \mid a \mid b \mid aSa \mid bSb$$

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Turing Machine

Definition 7.1 Turing Machines

A Turing machine (TM) is a 5-tuple $T = (Q, \Sigma, \Gamma, q_0, \delta)$, where

Q is a finite set of states. The two *halt* states h_a and h_r are not elements of Q.

Σ, the input alphabet, and Γ, the tape alphabet, are both finite sets, with $\Sigma \subseteq \Gamma$. The *blank* symbol Δ is not an element of Γ.

q_0, the initial state, is an element of Q.

δ is the transition function:

$$\delta : Q \times (\Gamma \cup \{\Delta\}) \rightarrow (Q \cup \{h_a, h_r\}) \times (\Gamma \cup \{\Delta\}) \times \{\mathrm{R}, \mathrm{L}, \mathrm{S}\}$$

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Turing Machines for Computation

- Given *n* find $\log_2 n$
- Given *n* and *m* find $\log_m n$
- Find difference between two numbers
- Given an array of numbers find its mean
- Given two arrays of numbers find mean of paired differences

Turing Machine and Semantics

- Find difference between two words
- Find **semantic** difference between two words, sentences, and documents

(Recall Marr's Levels of Analysis)

Models, Language and Language Model

- What do you mean by a **model** of a phenomenon?
- Is it feasible to have a **model** of a **language**?
- How can we define a **model** of a **language**?
- How can we define a **computational model** of a language?
- How can we evaluate a **computational model** of a language?

Statistical Models of NLP

- Why statistical models?
  - We know statistics, ML and DL for modeling operations on numbers
- Why probability?
  - We can model uncertainty
- Why not “logical models” of a language?

Flattening a meaning

- A word is equivalent to a number
- A sentence is a sequence of numbers
- A document is a sequence of sentences

- Apply (machine|deep) learning algorithms for *sequences* on sentences|documents

Validity of these assumptions?